

THE APPLICATION OF THE LICHTENSTEIN-GERSHGORIN INTEGRAL
EQUATION IN CONFORMAL MAPPING.

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Preface.

During the years 1959 and 1960 Mr. S.E. Christiansen, Mr. Ole Møller, Professor Hans Tornehave and the author have studied different methods from the theory of conformal mapping in order to find methods which were specially suited for numerical solution on digital computers¹⁾.

In accordance with several others²⁾ we found that the methods which may be called the integral equation methods were well suited for our purpose, and we decided that the Lichtenstein-Gershgorin integral equation method should be programmed for the Danish computer DASK³⁾.

In part I of this paper the Lichtenstein-Gershgorin integral equation is developed and some questions about the kernel and other questions about a special system of linear equations are discussed.

Part II contains a description of the program and some results obtained by numerical experiments.

1) Our work has been a part of a larger project sponsored by the Carlsberg Foundation. A report [1] containing the different papers from the members of the Carlsberg group will appear in 1962. (The numbers in brackets refer to the papers in the literature references.)

2) See Birkhoff, Young and Zarantonello [2].

3) S.E. Warschawski has discussed the theory of this method [3] and together with John Todd [4] given many computational results obtained by using this and other methods.

1. Introduction.

In the complex z -plane we consider a closed Jordan curve C with continuously turning tangent. Let D denote the interior of C and let $z_0 \in D$ and $z_1 \in C$.

From the mapping theorem of Riemann¹⁾ we know that there exists a unique analytic function $w = f(z)$ with the following properties:

- 1) $w = f(z)$ maps D conformally onto the circle $|w| < 1$,
- 2) $f(z_0) = 0$ and $f(z_1) = 1$.

If we extend the definition of $f(z)$ to C in such a way that $f(z)$ is continuous in $D+C$ we have established a one-to-one correspondence between the points of C and the points of $|w| = 1$. Let s denote the length of the arc on C from z_1 to $\zeta(s)$, then we know the angle $\theta(s) = \arg f(\zeta(s))$ as a function of s .

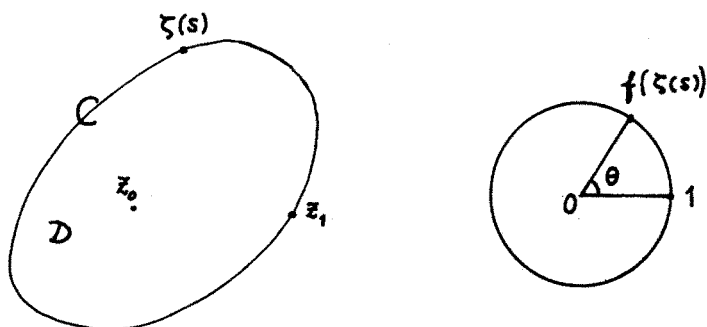


Figure 1.

From the integral theorem of Cauchy,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta \quad z \in D$$

we have

$$f(z) = \frac{1}{2\pi i} \int_0^L \frac{\cos \theta(s) + i \sin \theta(s)}{\zeta(s) - z} \frac{d\zeta}{ds} ds \quad (1)$$

where L is the length of C . Thus we can find $f(z)$ if we know θ as a function of s .

1) A proof of this theorem may be found in Neharis book [5] or in the little book [6] b Bieberbach.

On C_ϵ we have

$$|F(\zeta)| \leq M |\log \epsilon|.$$

Furthermore $|\zeta - z| > \frac{|z - z_0|}{2} = k$ when $\zeta \in C_\epsilon$ (and $\epsilon < \frac{|z - z_0|}{2}$).

$$\text{Hence } \left| \frac{1}{2\pi i} \int_{C_\epsilon} \frac{F(\zeta)}{\zeta - z} d\zeta \right| \leq \frac{M |\log \epsilon|}{2\pi k} \epsilon \cdot 2\pi,$$

which tends to zero with ϵ .

Now

$$\begin{aligned} \frac{1}{2\pi i} \int_{z_1 \text{ to } C_\epsilon} \frac{F(\zeta)}{\zeta - z} d\zeta + \frac{1}{2\pi i} \int_{C_\epsilon \text{ to } z_1} \frac{F(\zeta)}{\zeta - z} d\zeta &= \frac{1}{2\pi i} \int_{z_1 \text{ to } C_\epsilon} \frac{F(\zeta_+) - F(\zeta_-)}{\zeta - z} d\zeta = \\ &= \frac{1}{2\pi i} \int_{z_1 \text{ to } C_\epsilon} \frac{2\pi i}{\zeta - z} d\zeta. \end{aligned}$$

When $\epsilon \rightarrow 0$ the last integral tends to

$$\int_{z_1}^{z_0} \frac{d\zeta}{\zeta - z} = \left[\log(\zeta - z) \right]_{z_1}^{z_0} = -\log \frac{z_1 - z}{z_0 - z}.$$

With these results and the fact that $F(z) = i\theta$ on C we have (from (4))

$$F(z) = \frac{1}{2\pi} \int_C \frac{\theta d\zeta}{\zeta - z} - \log \frac{z_1 - z}{z_0 - z}. \quad (5)$$

Let $\zeta - z = r \cdot e^{i\varphi}$, $z_0 - z = r_0 e^{i\varphi_0}$ and $z_1 - z = r_1 e^{i\varphi_1}$.

Then $\varphi_1 - \varphi_0 = \beta$ is the angle from the line zz_0 to the line zz_1 (see figure 2).

Let α be the angle from the x-axis to the tangent to C at the point ζ , and let t denote the arc length of C , starting from z_1 (or from some other point).

Lastly we introduce the angle (n, r) from the interior normal to C (at ζ) to the line ζz .

Then we have

$$d\zeta = e^{i\alpha} dt$$

$$\alpha - \varphi = \frac{\pi}{2} - (n, r) \quad (\text{see figure 2}).$$

From (5) we obtain

$$F(z) = \frac{1}{2\pi} \int_C \frac{\Theta e^{i\alpha} dt}{r e^{i\varphi}} - \log \frac{r_1}{r_0} - i\beta = \frac{1}{2\pi} \int_C \frac{\Theta e^{\frac{i\pi}{2}} e^{-i(n,r)} dt}{r} - \log \frac{r_1}{r_0} - i\beta = \frac{i}{2\pi} \int_C \frac{\Theta e^{-i(n,r)}}{r} dt - \log \frac{r_1}{r_0} - i\beta. \quad (6)$$

From (6) we obtain - by separating the real and imaginary parts

$$\log \rho = \frac{1}{2\pi} \int_C \frac{\Theta \sin(n,r)}{r} dt - \log \frac{r_1}{r_0} \quad (7)$$

$$\Theta = \frac{1}{2\pi} \int_C \frac{\Theta \cos(n,r)}{r} dt - \beta \quad (8)$$

It follows that $\log \rho$ and Θ are determined as soon as the boundary values of the function Θ are known. As we shall see, it is possible to deduce an integral equation for the boundary values of Θ from (8). To obtain this equation z in (8) must tend to a point $\zeta(s)$ on C . In this way we obtain

$$\Theta(s) = \frac{1}{2\pi} \left[\int_C \frac{\Theta(t) \cos(n_t, r_{st})}{r_{st}} dt + \Theta(s) \cdot \pi \right] - \beta(s) \quad (9)$$

from which we have the Lichtenstein-Gershgorin Integral equation

$$\Theta(s) = \frac{1}{\pi} \int_C \frac{\Theta(t) \cos(n_t, r_{st})}{r_{st}} dt - 2\beta(s). \quad (10)$$

Before we prove (9) we remark that the kernel $\frac{\cos(n_t, r_{st})}{r_{st}}$

in (9) is not defined if $t = s$. This singularity is not essential, since the limit of the kernel exists when $t \rightarrow s$. This will be proved later. A further remark is, that

$\int_C \frac{\Theta(t) \cos(n_t, r_{zt})}{r_{zt}} dt$ is not continuous as a function of z ,

when z tends to a boundary point $\zeta(s)$. We will prove, however, that the function

$$\int_C [\Theta(t) - \Theta(s)] \frac{\cos(n_t, r_{zt})}{r_{zt}} dt \quad (11)$$

has the limit

$\int_C [\theta(t) - \theta(s)] \frac{\cos(n_t, r_{st})}{r_{st}} dt$ as z tends to $\zeta(s)$; that is to every $\varepsilon > 0$ we can find a δ such that

$$\left| \int_C [\theta(t) - \theta(s)] \left[\frac{\cos(n_t, r_{zt})}{r_{zt}} - \frac{\cos(n_t, r_{st})}{r_{st}} \right] dt \right| < \varepsilon \text{ when } (12)$$

$$|z - \zeta(s)| < \delta.$$

Let C be divided into C_1

and C_2 where $\zeta(s) \in C_1$,

and let I_1 and I_2 denote

the integrals of the function from (12) along C_1 and C_2 , respectively.

Let $\max_{\zeta(t) \in C} |\theta(t) - \theta(s)| = M$

and let L be the length of C . Since $\zeta(s) \notin C_2$ it follows, that we can find a δ such that

$$\left| \frac{\cos(n_t, r_{zt})}{r_{zt}} - \frac{\cos(n_t, r_{st})}{r_{st}} \right| < \frac{\varepsilon}{ML}$$

for all $|z - \zeta(s)| < \delta$ and all $\zeta(t) \in C_2$, that is $|I_2|$ can be made arbitrarily small.

We introduce the angle ω from the x -axis to the line from z to $\zeta(t)$. (See figure 3). ω is a function of z and t .

For a fixed z we have

$$d\omega = \frac{\cos(n_t, r_{zt})}{r_{zt}} dt \quad (13)$$

from which we deduce the result

$$\int_C \frac{\cos(n_t, r_{zt})}{r_{zt}} dt = \int_C d\omega = 2\pi \quad (z \in D) \quad (14)$$

Let p be an integer with the property that no line in the z -plane intersects C more than p times; thus

$$\int_C \frac{|\cos(n_t, r_{zt})|}{r_{zt}} dt \leq p \cdot 2\pi. \quad (15)$$

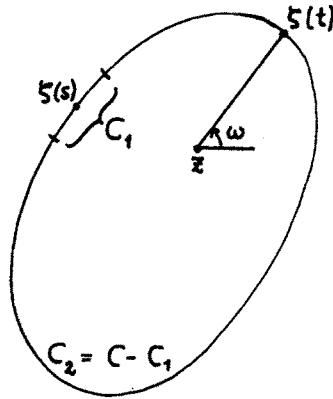


Figure 3.

If $z = \zeta(s) \in C$ we may deduce

$$\int_C \frac{\cos(n_t, r_{st})}{r_{st}} dt = \pi \quad (\zeta(s) \in C) \quad (16)$$

We remark that (15) is valid also in this case. Let us return to the integral I_1 , and let $\max_{\zeta(t) \in C_1} |\theta(t) - \theta(s)| = M_1$;

it follows, that $M_1 \rightarrow 0$ when the length of C_1 tends to zero.

Now

$$\begin{aligned} |I_1| &\leq M_1 \int_{C_1} \left| \frac{\cos(n_t, r_{zt})}{r_{zt}} - \frac{\cos(n_t, r_{st})}{r_{st}} \right| dt \leq \\ &\leq M_1 \int_C \left| \frac{\cos(n_t, r_{zt})}{r_{zt}} - \frac{\cos(n_t, r_{st})}{r_{st}} \right| dt \leq M_1 4\pi p; \end{aligned}$$

that is $|I_1|$ can be made arbitrarily small, and we have proved that

$$\lim_{z \rightarrow \zeta(s)} \int_C [\theta(t) - \theta(s)] \frac{\cos(n_t, r_{zt})}{r_{zt}} dt = \int_C [\theta(t) - \theta(s)] \frac{\cos(n_t, r_{st})}{r_{st}} dt$$

From this formula we obtain - by means of (14) and (16) -

$$\begin{aligned} \lim_{z \rightarrow \zeta(s)} \int_C \theta(t) \frac{\cos(n_t, r_{zt})}{r_{zt}} dt - \theta(s) \cdot 2\pi = \\ = \int_C \theta(t) \frac{\cos(n_t, r_{st})}{r_{st}} dt - \theta(s) \cdot \pi \end{aligned}$$

which immediately shows that (9) follows from (8).

3. Polar coordinates.

In (10) s and t denote the length of the arcs from a fixed point on C to the points $\zeta(s)$ and $\zeta(t)$, respectively. If C is given in the form $x = x(s)$, $y = y(s)$ it is possible

to find the kernel $\frac{\cos(n_t, r_{st})}{r_{st}}$ directly as a function of s and t .

Usually the functions $x = x(s)$ and $y = y(s)$ are not known

beforehand and we have therefore chosen another parameter. Let C be given by an equation

$$r = r(v),$$

in polar coordinates with z_0 as pole. We remark that this is only possible if D is star-shaped. Instead of s and t we will use the polar angles v and u as parameters.

Using v and u in (10) we obtain

$$\theta(v) = \frac{1}{\pi} \int_0^{2\pi} \theta(u) \frac{\cos(n_u, r_{uv})}{r_{uv}} \frac{dt}{du} du - 2\beta(v), \quad (17)$$

which may be written

$$\theta(v) = \int_0^{2\pi} K(v, u) \theta(u) du - 2\beta(v), \quad (18)$$

with the kernel

$$K(v, u) = \frac{\cos(n_u, r_{uv})}{\pi r_{uv}} \frac{dt}{du} \quad (19)$$

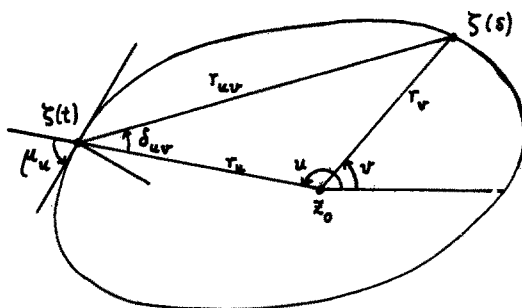


Figure 4.

From (19) we obtain (see figure 4)

$$\begin{aligned} K(v, u) &= \frac{\sin(\mu_u - \delta_{uv})}{\pi r_{uv}} \frac{dt}{du} = \frac{\sin(\mu_u - \delta_{uv})}{\pi r_{uv}} \cdot \frac{r_u}{\sin \mu_u} = \\ &= \frac{r_u \cos \delta_{uv} - r_u \cot \mu_u \sin \delta_{uv}}{\pi r_{uv}} = \end{aligned}$$

$$= \frac{r_u [r_{uv}^2 + r_u^2 - r_v^2]}{\pi 2 r_{uv}^2 r_u} - \frac{r_u r_v \sin(u-v) \frac{dr}{du}}{\pi r_{uv}^2 \cdot r_u}$$

and finally

$$K(v, u) = \frac{r_u [r_u - r_v \cos(u-v)] - r_v \sin(u-v) \frac{dr}{du}}{\pi [r_u^2 + r_v^2 - 2 r_u r_v \cos(u-v)]} \quad (20)$$

The kernel is singular when $u = v$, but this singularity is not essential since $\lim_{u \rightarrow v} K(v, u)$ exists.

$$K(v, v) = \lim_{u \rightarrow v} K(v, u) = \frac{r_v^2 + 2 \left(\frac{dr}{dv} \right)^2 - r_v \frac{d^2 r}{dv^2}}{2\pi \left[r_v^2 + \left(\frac{dr}{dv} \right)^2 \right]} \quad (21)$$

4. Numerical determination of $\theta(v)$.

The integral equation

$$\theta(v) = \int_0^{2\pi} K(v, u) \theta(u) du - 2\beta(v) \quad (22)$$

from which we want to find $\theta(v)$ is solved numerically in the following manner: We choose the following values of v ; $v_0, v_1, \dots, v_N$ in the interval 0 to 2π , and try to determine the values

$$\theta(v_0), \theta(v_1), \dots, \theta(v_N).$$

The value $\theta(v)$ where $v \neq v_i$ ($i = 0, 1, \dots, N$) must be found by interpolation.

From (16) it follows that

$$\theta(v_i) = \int_0^{2\pi} K(v_i, u) \theta(u) du - 2\beta(v_i) \quad (i = 0, 1, \dots, N) \quad (23)$$

The integral in (23) is now replaced by a linear combination of $\theta(v_0), \theta(v_1), \dots, \theta(v_N)$ and we get a system of linear equations

$$\tilde{\theta}(v_i) = \sum_{j=0}^N A_j K(v_i, v_j) \tilde{\theta}(v_j) - 2\beta(v_i) \quad (i = 0, 1, \dots, N) \quad (24)$$

The values v_i and the coefficients A_j depend on the formula

of quadrature which has been used. This formula is not exact and we have used the notation $\tilde{\Theta}(v_i)$ instead of $\Theta(v_i)$.

In connection with this procedure several questions must be answered.

- 1) Has the system (24) a solution? If yes
- 2) Which method should be used to find the solution (or the solutions)?
- 3) What can be said about the discrepancy between $\tilde{\Theta}(v_i)$ and $\Theta(v_i)$?

We shall try to answer these questions in the following.

5. The trapezoid formula.

The quadrature formula which has been used is the trapezoid formula with correction. This formula may be obtained from the Euler-Maclaurin formula¹⁾. Let $f^{(4)}(x)$ be continuous in the interval $x_0 \leq x \leq x_N$ and let $x_i = x_0 + h \cdot i$ ($i = 0, 1, 2, \dots, N$). Then

$$\int_{x_0}^{x_N} f(x) dx = h \left(\frac{1}{2} f(x_0) + f(x_1) + \dots + f(x_{N-1}) + \frac{1}{2} f(x_N) \right) - \frac{B_2}{2!} h^2 [f'(x_N) - f'(x_0)] + E; \quad (25)$$

$$E = -N \frac{B_4}{4!} h^5 f^{(4)}(\xi) \quad x_0 < \xi < x_N \quad (26)$$

B_2 and B_4 are the Bernoulli numbers $\frac{1}{6}$ and $-\frac{1}{30}$, respectively. In our case $f(x)$ has to be replaced by the function

$$K(v, u) \cdot \Theta(u).$$

The interval runs from 0 to 2π and $v_i = \frac{2\pi}{N} i$ ($i = 0, 1, 2, \dots, N$). Then

$$\int_0^{2\pi} K(v, u) \Theta(u) du = \frac{2\pi}{N} \left[\frac{1}{2} K(v, 0) \Theta(0) + K(v, \frac{2\pi}{N}) \Theta(\frac{2\pi}{N}) + \dots + \frac{1}{2} K(v, 2\pi) \Theta(2\pi) \right] - \frac{1}{12} \left(\frac{2\pi}{N} \right)^2 \left[\frac{\partial}{\partial u} (K(v, u) \Theta(u)) \right]_{u=0}^{u=2\pi} + E, \quad (27)$$

$$E = N \frac{1}{720} \left(\frac{2\pi}{N} \right)^5 \frac{\partial^4}{\partial u^4} (K(v, \xi) \Theta(\xi)); \quad 0 < \xi < 2\pi \quad (28)$$

1) See for example the book by Hildebrand [8].

Now the kernel $K(v, u)$ is periodic with the period 2π and $\theta(u+2\pi) = \theta(u) + 2\pi$. Hence

$$\frac{1}{2}K(v, 2\pi)\theta(2\pi) = \frac{1}{2}K(v, 0)[\theta(0)+2\pi], \quad (29)$$

and we obtain

$$\begin{aligned} \int_0^{2\pi} K(v, u)\theta(u)du &= \frac{2\pi}{N} \left[K(v, 0)[\theta(0)+\pi] + K(v, \frac{2\pi}{N})\theta(\frac{2\pi}{N}) + \dots + \right. \\ &\quad \left. + K(v, \frac{2\pi}{N}(N-1)) \right] - \frac{1}{12} \left(\frac{2\pi}{N} \right)^2 \left[\frac{\partial}{\partial u} (K(v, u)\theta(u)) \right]_{u=0}^{u=2\pi} + E \end{aligned} \quad (30)$$

$$\text{Now } \frac{\partial}{\partial u} (K(v, u)\theta(u)) = K(v, u) \frac{d\theta(u)}{du} + \frac{\partial}{\partial u} (K(v, u))\theta(u).$$

Since $K(v, 2\pi) = K(v, 0)$; $\theta(2\pi) = \theta(0) + 2\pi$ and $\theta'(2\pi) = \theta'(0)$,

we have

$$\left[\frac{\partial}{\partial u} (K(v, u)\theta(u)) \right]_{u=0}^{u=2\pi} = 2\pi \left[\frac{\partial}{\partial u} K(v, u) \right]_{u=0} \quad (31)$$

Let us assume that the curve C intersects the polar axis orthogonally. Then

$$\left[\frac{dr(u)}{du} \right]_{u=0} = 0 \quad (32)$$

By differentiation with respect to u of (20) and taking (32) into account we then obtain

$$\begin{aligned} 2\pi \left[\frac{\partial K(v, u)}{\partial u} \right]_{u=0} &= \frac{2r_v \sin v \left[\frac{d^2 r(u)}{du^2} \right]_{u=0}}{r_0^2 + r_v^2 - 2r_0 r_v \cos v} + \\ &\quad + \frac{2r_0 r_v \sin v (r_0^2 - r_v^2)}{(r_0^2 + r_v^2 - 2r_0 r_v \cos v)^2} \quad \text{provided } v \neq 0. \end{aligned} \quad (33)$$

If $v = 0$ we have (by use of (32))

$$\begin{aligned} 2\pi \left[\frac{\partial K(0, u)}{\partial u} \right]_{u=0} &= 2\pi \lim_{u \rightarrow 0} \frac{K(0, u) - K(0, 0)}{u} \\ &= 2\pi \lim_{u \rightarrow 0} \frac{1}{u} \left[\frac{r_u^2 - r_0^2 \cos u - r_0 \sin u \frac{dr(u)}{du}}{\pi(r_u^2 + r_0^2 - 2r_0 r_u \cos u)} \right] - \end{aligned}$$

$$- \frac{r_0^2 - r_0 \left[\frac{d^2 r(v)}{dv^2} \right]_{v=0}}{2\pi r_0^2} \Bigg]_{u \rightarrow 0} = \lim_{u \rightarrow 0} \frac{T(u)}{N(u)} .$$

Now

$$T(0) = N(0) = 0 ; T'(0) = N'(0) = 0 \text{ and } T''(0) = N''(0) = 0 ,$$

but

$$T'''(0) = -4 r_0^3 \left[\frac{d^3 r(u)}{du^3} \right]_{u=0} \text{ and } N'''(0) = 6 r_0^4 ,$$

hence

$$2\pi \left[\frac{\partial K(0, u)}{\partial u} \right]_{u=0} = - \frac{2}{3 r_0} \left[\frac{d^3 r(u)}{du^3} \right]_{u=0} \quad (34)$$

By means of (30) and (31) we have from (22)

$$\begin{aligned} \Theta(v) &= \frac{2\pi}{N} K(v, 0) (\Theta(0) + \pi) + \sum_{j=1}^{N-1} \frac{2\pi}{N} K(v, \frac{2\pi}{N} j) \Theta(\frac{2\pi}{N} j) - \\ &- \frac{1}{12} \left(\frac{2\pi}{N} \right)^2 2\pi \left[\frac{\partial K(v, u)}{\partial u} \right]_{u=0} - 2\beta(v) + E_v , \end{aligned} \quad (35)$$

or

$$\begin{aligned} \Theta(v) &= \sum_{j=0}^{N-1} \frac{2\pi}{N} K(v, \frac{2\pi}{N} j) \Theta(\frac{2\pi}{N} j) - 2\beta(v) - \\ &- \frac{1}{12} \left(\frac{2\pi}{N} \right)^2 2\pi \left[\frac{\partial K(v, u)}{\partial u} \right]_{u=0} + \frac{2\pi}{N} K(v, 0) \cdot \pi + E_v . \end{aligned} \quad (36)$$

Putting $v_0 = 0$, $v_1 = \frac{2\pi}{N}$, ... $v_{N-1} = \frac{2\pi}{N}(N-1)$ in (36) and

changing the notation so that $K(\frac{2\pi}{N} i, \frac{2\pi}{N} j) = K_{ij}$;

$\Theta(\frac{2\pi}{N} i) = \Theta_i$; $\beta(\frac{2\pi}{N} i) = \beta_i$; $E_{v_i} = E_i$ and $\frac{2\pi}{N} = h$ we obtain

the following system of linear equations:

$$\begin{aligned} \Theta_i &= \sum_{j=0}^{N-1} h K_{ij} \Theta_j - 2\beta_i - \frac{\pi}{6} h^2 \left[\frac{\partial K(v_i, u)}{\partial u} \right]_{u=0} + \\ &+ h K_{i0} \cdot \pi + E_i \quad (i = 0, 1, 2, \dots, N-1) \end{aligned} \quad (37)$$

or

$$\tilde{\theta}_i = \sum_{j=0}^{N-1} h K_{ij} \tilde{\theta}_j + \alpha_i \quad (i = 0, 1, \dots, N-1) \quad (38)$$

where

$$\alpha_i = -2\beta_i - \frac{\pi}{6} h^2 \left[\frac{\partial K(v_i, u)}{\partial u} \right]_{u=0} + h K_{i0} \cdot \pi \quad (39)$$

6. The solution of the system of linear equations.

From (38) we obtain the following coefficient matrix

$$A_{NN} = \begin{pmatrix} 1 - h K_{00} & -h K_{01} & \dots & -h K_{0,N-1} \\ -h K_{10} & 1 - h K_{11} & \dots & -h K_{1,N-1} \\ \vdots & \vdots & \ddots & \vdots \\ -h K_{N-1,0} & -h K_{N-1,1} & \dots & 1 - h K_{N-1,N-1} \end{pmatrix} \quad (40)$$

We want to show that

$$\det A_{NN} = 0. \quad (41)$$

From (16) we have that

$$\int_0^{2\pi} K(v, u) du = \int_0^{2\pi} \frac{\cos(n_u, r_{uv})}{\pi r_{uv}} \frac{dt}{du} du = 1.$$

Then

$$\begin{aligned} 1 &= \int_0^{2\pi} K(v, u) du = \frac{2\pi}{N} \left[\frac{1}{2} K(v, 0) + K(v, \frac{2\pi}{N}) + \dots + \right. \\ &\quad \left. + K(v, \frac{2\pi}{N}(N-1)) + \frac{1}{2} K(v, 2\pi) \right] - \frac{1}{12} \left(\frac{2\pi}{N} \right)^2 \left[\frac{\partial}{\partial u} K(v, u) \right]_{u=0}^{u=2\pi} + \\ &\quad + E^*, \end{aligned} \quad (42)$$

with the error

$$E^* = N \cdot \frac{1}{720} \left(\frac{2\pi}{N} \right)^5 \left[\frac{\partial^4}{\partial u^4} (K(v, u)) \right]_{u=\xi}; \quad 0 < \xi < 2\pi.$$

Neglecting this error in (42) and using that the kernel

$K(v, u)$ is periodic with the period 2π we obtain - in the usual notation -

$$\sum_{j=0}^{N-1} h K_{ij} = 1 \quad (i = 0, 1, \dots, N-1) \quad (43)$$

By adding the columns in (40) and by the use of (43) we obtain a zero-column; that is $\det A_{NN} = 0$.

It follows, that (38) has either infinitely many solutions or (38) cannot be solved.

We remark, that as soon as we have a solution

$$(\tilde{\theta}_0; \tilde{\theta}_1; \dots; \tilde{\theta}_{N-1})$$

then

$$(\tilde{\theta}_0+K; \tilde{\theta}_1+K; \dots; \tilde{\theta}_{N-1}+K)$$

will also be a solution. This follows immediately by use of (43). Furthermore we can say, that there is nothing surprising in the fact that there are infinitely many solutions to (38), since we have not used the condition that the point $z = z_1$ should be mapped in the point $w = 1$.

Let the line $z_0 z_1$ be the polar-axis. Then $\tilde{\theta}(0) = \tilde{\theta}_0$ should be zero.

Putting $\tilde{\theta}_0 = 0$ in (38) it follows that $(\tilde{\theta}_1, \tilde{\theta}_2, \dots, \tilde{\theta}_{N-1})$ must satisfy the equations

$$\begin{aligned} \tilde{\theta}_1 &= h K_{11} \tilde{\theta}_1 + h K_{12} \tilde{\theta}_2 + \dots + h K_{1,N-1} \tilde{\theta}_{N-1} + \alpha_1 \\ \tilde{\theta}_2 &= h K_{21} \tilde{\theta}_1 + h K_{22} \tilde{\theta}_2 + \dots + h K_{2,N-1} \tilde{\theta}_{N-1} + \alpha_2 \\ &\vdots \\ \tilde{\theta}_{N-1} &= h K_{N-1,1} \tilde{\theta}_1 + h K_{N-1,2} \tilde{\theta}_2 + \dots + \\ &\quad + h K_{N-1,N-1} \tilde{\theta}_{N-1} + \alpha_{N-1} \end{aligned} \quad (44)$$

or in matrix notation

$$\theta = K \theta + \alpha \quad (45)$$

We want to solve (45) by an iteration process. Before do-

ing so, we mention some theorems about matrices¹⁾.

Let K^n be the n th power of the matrix K . In order that $K^n \rightarrow 0$ as $n \rightarrow \infty$, it is necessary and sufficient that all the eigenvalues of K are of modulus less than unity.

Proof. If K can be brought into a diagonal form, $K = U\Lambda U^{-1}$, with

$$\Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ & & \ddots \\ 0 & & & \lambda_{N-1} \end{pmatrix}$$

where $\lambda_1, \lambda_2, \dots, \lambda_{N-1}$ are the eigenvalues of K , we have that $K^n = U\Lambda^n U^{-1}$, where

$$\Lambda^n = \begin{pmatrix} \lambda_1^n & & 0 \\ & \lambda_2^n & \\ & & \ddots \\ 0 & & & \lambda_{N-1}^n \end{pmatrix}$$

In order that $K^n \rightarrow 0$ it is necessary and sufficient that $\Lambda^n \rightarrow 0$, and the theorem follows at once. If K cannot be brought into diagonal form, K can be brought into Jordan canonical form²⁾ and the proof can be established.

Next we prove the following theorem:

The series

$$I + K + K^2 + K^3 + \dots + K^n + \dots \quad (46)$$

converges if $K^n \rightarrow 0$ as $n \rightarrow \infty$. The sum of (46) will then be $(I - K)^{-1}$.

Proof. Let $K^n \rightarrow 0$ as $n \rightarrow \infty$. Since all the eigenvalues are of modulus less than unity we conclude that

$\det(I - K) \neq 0$, and therefore $(I - K)^{-1}$ exists.

1) See for example the book by Faddeeva [9].

2) The book [10] by Aitken contains a proof of this theorem.

From the identity

$$I + K + \dots + K^n = (I - K)^{-1} - K^{n+1}(I - K)^{-1}$$

it follows at once, that

$$I + K + \dots + K^n + \dots = (I - K)^{-1}, \quad (47)$$

since $K^{n+1} \rightarrow 0$.

To obtain a solution of (45) by iteration we define the vector θ_n by

$$\theta_n = K \theta_{n-1} + \alpha \quad (n = 1, 2, 3, \dots) \quad (48)$$

θ_0 may be an arbitrary vector. From (48) we have

$$\theta_n = K^n \theta_0 + (I + K + K^2 + \dots + K^{n-1}) \alpha \quad (49)$$

Provided that the eigenvalues of K are of modulus less than unity we have - as a consequence of the theorem just proved - that

$$\lim_{n \rightarrow \infty} \theta_n = (I - K)^{-1} \alpha \quad (50)$$

In many cases the theorems are of little help since one does not know the eigenvalues of K . We shall therefore establish another theorem rendering $K^n \rightarrow 0$ as $n \rightarrow \infty$.

Let K_{ij}^n denote the (i, j) -element of K^n . We introduce the number defined by

$$\|K\| = \max_i \sum_{j=1}^{N-1} |K_{ij}^1| \quad (51)$$

By induction follows at once that

$$\max_{ij} |K_{ij}^n| \leq \|K\|^n \quad (52)$$

from which we have, that

$$K^n \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ provided } \|K\| < 1. \quad (53)$$

This theorem is of great use in our case, at least if the domain D , which will be mapped on the unit circle, is convex. In this case all the elements hK_{ij} of the matrix in (44) are positive, since the kernel is a positive function

as soon as D is a convex domain.

Furthermore we have from (43) that

$$\sum_{j=1}^{N-1} h K_{ij} = 1 - h K_{i0}$$

All the numbers

$$1 - h K_{i0} \quad (i = 1, 2, \dots, N-1)$$

are positive and less than unity; that is $\|K\| < 1$ and it follows that the iteration process will converge.

We remark that in other cases, when the domain D is only star-shaped, one can try to show the convergence of the iteration process by means of another norm instead of

$$\|K\| = \max_i \sum_{j=1}^{N-1} |K_{ij}^1|.$$

We finish this section by stating a result from the corresponding continuous problem.

Let $\theta_{n+1}(v)$ be defined by

$$\theta_{n+1}(v) = \int_0^{2\pi} K(v, u) \theta_n(u) du - 2\beta(v).$$

Then the sequence

$$\theta_1(v); \theta_2(v); \dots, \theta_n(v), \dots$$

converges uniformly to a solution $\theta(v)$ of

$$\theta(v) = \int_0^{2\pi} K(v, u) \theta(u) du - 2\beta(v).$$

For a proof of this theorem see the paper [3] by Warshawski.

II. Numerical experiments.

7. The program.

The program, which I prepared for the Danish computer DASK, should be able to handle an arbitrary contour C , given by an equation $r = r(v)$. In the computations we use the values

$$r_i = r(v_i) \text{ with } v_i = \frac{2\pi}{N} i \quad (i = 0, 1, \dots, N-1).$$

Furthermore the values

$$\frac{dr(v_i)}{dv} \quad \text{and} \quad \frac{d^2r(v_i)}{dv^2}$$

should be known in order to compute the coefficients K_{ij} from the formulas (20) and (21). From a practical point of view it would be very important if these values might be computed by numerical differentiation from the values r_i . In this case it would not be necessary to know the analytic expressions for the functions $r(v)$, $\frac{dr}{dv}$ and $\frac{d^2r}{dv^2}$.

In order to see the influence of the numerical differentiation some of the computations were carried out twice,

first with the values of $\frac{dr(v_i)}{dv}$ and $\frac{d^2r(v_i)}{dv^2}$ obtained from

the analytical expressions of $\frac{dr}{dv}$ and $\frac{d^2r}{dv^2}$ and then repeated with the values found by numerical differentiation.

We were also interested in the importance of the correction term in the trapezoid formula used by the numerical quadrature, and some of the computations were carried out both with and without this correction.

The number of equations to be solved by the iteration process were limited by the memory capacity of DASK. I have chosen $N = 60$. The following tables then contain the values of θ corresponding to the values

$$0^\circ, 6^\circ, 12^\circ, \dots, 354^\circ \text{ of } v.$$

As starting vector θ_0 in the iteration process I have chosen the values

$$\frac{2\pi}{60}, 2\frac{2\pi}{60}, 3\frac{2\pi}{60}, \dots, 59\frac{2\pi}{60}.$$

The iteration was carried out until

$$\max_i |\theta_i^{n+1} - \theta_i^n| < 10^{-8}.$$

8. Mapping of a circle.

In order to check the program the computations were carried out with a circle in which the point z_0 was chosen at a distance b from the center of the circle (figure 5). Let a denote the radius of the circle. Then we have

$$r = b \cos v + \sqrt{a^2 - b^2 \sin^2 v}. \quad (54)$$

The different values of $\frac{dr}{dv}$ and $\frac{d^2r}{dv^2}$ were found by numerical differentiation by means of the formula

$$\begin{aligned} \frac{dr(v_i)}{dv} \approx \frac{1}{h} & \left[\frac{3}{4}(r_{i+1} - r_{i-1}) - \frac{3}{20}(r_{i+2} - r_{i-2}) + \right. \\ & \left. + \frac{1}{60}(r_{i+3} - r_{i-3}) \right]. \end{aligned} \quad (55)$$

The results are given in the following tables 1 and 2.

In this case it is easy to find the exact expression for the function $\Theta(v)$.

We first show that

$$w = \frac{az}{bz + a^2 - b^2}$$

is the mapping function of the domain. We have $z_0 = 0$ and $z_1 = a+b$, and

from (56) follows that $w(0) = 0$ and $w(a+b) = 1$. Let $z = r e^{iv}$ be a point on C . Then

$$w = \frac{a r e^{iv}}{b r e^{iv} + a^2 - b^2} \quad \text{and} \quad \bar{w} = \frac{a r e^{-iv}}{b r e^{-iv} + a^2 - b^2}$$

It follows that $w \bar{w} = 1$ i.e. $|w| = 1$.

Putting $w = e^{i\Theta}$ we obtain

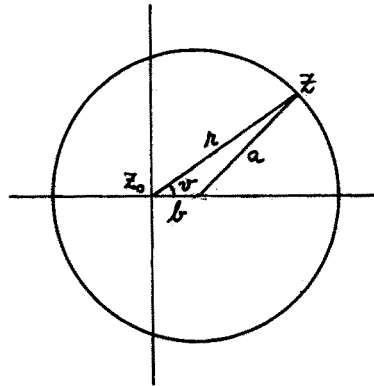


Figure 5.

$$e^{i\theta} = \frac{are^{iv}}{bre^{iv} + a^2 - b^2} \quad (57)$$

from which we deduce the result

$$\operatorname{tg} \theta = \frac{(a^2 - b^2) \sin v}{br + (a^2 - b^2) \cos v} \quad (58)$$

A special program was prepared to find θ directly from (58) and thus it was possible to check the values found by the general method.

Table 1.

Mapping of a circle $\frac{b}{a} = 0.2$
 The general method Numerical differentiation
 $\theta_i^0 = \frac{2\pi}{60} i$ Number of iterations: 2

v^0	θ	v^0	θ	v^0	θ	v^0	θ	v^0	θ
0	0.000000	6	0.083813	12	0.167845	18	0.252316	24	0.337442
30	0.423431	36	0.510489	42	0.598809	48	0.688576	54	0.779960
60	0.873115	66	0.968176	72	1.065260	78	1.164457	84	1.265837
90	1.369438	96	1.475276	102	1.583337	108	1.693578	114	1.805934
120	1.920312	126	2.036597	132	2.154653	138	2.274326	144	2.395445
150	2.517826	156	2.641276	162	2.765591	168	2.890559	174	3.015966
180	3.141593	186	3.267220	192	3.392626	198	3.517595	204	3.641909
210	3.765359	216	3.887741	222	4.008860	228	4.128532	234	4.246588
240	4.362873	246	4.477251	252	4.589607	258	4.699849	264	4.807909
270	4.913747	276	5.017349	282	5.118728	288	5.217926	294	5.315009
300	5.410071	306	5.503225	312	5.594609	318	5.684376	324	5.772696
330	5.859754	336	5.945744	342	6.030869	348	6.115340	354	6.199373

The control by the special program gave the same results.

Table 2.

Mapping of a circle

$$\frac{b}{a} = 0.6$$

The general method

Numerical differentiation

$$\theta_i^0 = \frac{2\pi}{60} i$$

Number of iterations: 3

v^0	θ	v^0	θ	v^0	θ	v^0	θ	v^0	θ
0	0.000000	6	0.041962	12	0.084367	18	0.127670	24	0.172347
30	0.218906	36	0.267894	42	0.319908	48	0.375593	54	0.435645
60	0.500797	66	0.571794	72	0.649359	78	0.734146	84	0.826678
90	0.927296	96	1.036118	102	1.153026	108	1.277678	114	1.409551
120	1.547993	126	1.692281	132	1.841669	138	1.995423	144	2.152850
150	2.313301	156	2.476182	162	2.640945	168	2.807081	174	2.974115
180	3.141593	186	3.309071	192	3.476106	198	3.642242	204	3.807004
210	3.969884	216	4.130335	222	4.287760	228	4.441513	234	4.590900
240	4.735187	246	4.873631	252	5.005507	258	5.130162	264	5.247072
270	5.355895	276	5.456511	282	5.549041	288	5.633825	294	5.711390
300	5.782386	306	5.847538	312	5.907591	318	5.963277	324	6.015291
330	6.064279	336	6.110838	342	6.155515	348	6.198819	354	6.241224

The control showed that the error never exceeded $5 \cdot 10^{-6}$.

9. Mapping of ellipses.

In the numerical experiments with ellipses z_0 was chosen at the origin and the polar axis along the major axis.

The function $r = r(v)$ is given by the expression

$$r = \frac{b}{\sqrt{1 - e^2 \cos^2 v}} \quad (59)$$

Four experiments were carried out with the same ellipse in which the ratio $\frac{a}{b}$ was chosen equal to 1.2.

In the first case we used the general method which we have developed. The values of $\frac{dr(v_i)}{dv}$ and $\frac{d^2r(v_i)}{dv^2}$ were found by

means of the formulas obtained by differentiation of (59) once and twice, respectively.

The values of θ are given in table 3.

Table 3.

Mapping of an ellipse $\frac{a}{b} = 1.2$
 The general method Analytical differentiation
 $\theta_i^0 = \frac{2\pi}{60} i$ Number of iterations: 4

v°	θ	v°	θ	v°	θ	v°	θ	v°	θ
0	0.000000	6	0.084374	12	0.169850	18	0.257554	24	0.348306
30	0.442701	36	0.541069	42	0.643498	48	0.749867	54	0.859899
60	0.973198	66	1.089291	72	1.207656	78	1.327741	84	1.448980
90	1.570796	96	1.692612	102	1.813851	108	1.933936	114	2.052301
120	2.168395	126	2.281693	132	2.391725	138	2.498095	144	2.600523
150	2.698891	156	2.793286	162	2.884038	168	2.971743	174	3.057246
180	3.141593	186	3.225939	192	3.311442	198	3.399147	204	3.489899
210	3.584294	216	3.682662	222	3.785091	228	3.891460	234	4.001492
240	4.114791	246	4.230884	252	4.349249	258	4.469334	264	4.590573
270	4.712389	276	4.834205	282	4.955444	288	5.075529	294	5.193894
300	5.309988	306	5.423286	312	5.533318	318	5.639688	324	5.742116
330	5.840484	336	5.934879	342	6.025631	348	6.113336	354	6.198839

The values of θ corresponding to $v = 90^\circ$, $v = 180^\circ$ and $v = 270^\circ$ are correct to 7 figures (they are $= \frac{\pi}{2}$, π and $\frac{3\pi}{2}$).

Other values were checked by Mr. Ole Møller who has programmed another method for numerical conformal mapping. A paper about this method will appear in [1].

Furthermore some values were checked by interpolation in a table from the paper [4] by Warschawski and Todd. In all cases we found that the values in our table 3 were

correct.

In order to test the influence of the numerical differentiation, the experiment was repeated using formula (55).

Table 4.

Mapping of an ellipse $\frac{a}{b} = 1.2$
 The general method Numerical differentiation
 $\theta_i^0 = \frac{2\pi}{60} i$ Number of iterations: 5

v^0	θ	v^0	θ	v^0	θ	v^0	θ	v^0	θ
0	0.000000	6	0.084342	12	0.169847	18	0.257552	24	0.348305
30	0.442701	36	0.541069	42	0.643497	48	0.749867	54	0.859898
60	0.973197	66	1.089290	72	1.207656	78	1.327741	84	1.448980
90	1.570796	96	1.692612	102	1.813851	108	1.933936	114	2.052301
120	2.168394	126	2.281693	132	2.391726	138	2.498096	144	2.600526
150	2.698896	156	2.793290	162	2.884041	168	2.971741	174	3.057239
180	3.141584	186	3.225933	192	3.311440	198	3.399150	204	3.489905
210	3.584300	216	3.682667	222	3.785093	228	3.891461	234	4.001492
240	4.114790	246	4.230883	252	4.349249	258	4.469334	264	4.590573
270	4.712389	276	4.834206	282	4.955444	288	5.075529	294	5.193894
300	5.309987	306	5.423287	312	5.533320	318	5.639692	324	5.742124
330	5.840495	336	5.934890	342	6.025638	348	6.113335	354	6.198830

The difference between corresponding numbers in table 3 and table 4 does not exceed $11 \cdot 10^{-6}$ in absolute value.

The last two experiments in this section show the importance of the correction term in the trapezoid formula. Table 5 contains the results corresponding to table 3, and table 6 the results corresponding to table 4. The only difference is that the values in tables 5 and 6 have been computed using the trapezoid formula without correction term.

Table 5.

Mapping of an ellipse

$$\frac{a}{b} = 1.2$$

Trapezoid rule without correction

Analytical differentiation

$$\theta_i^0 = \frac{2\pi}{60} i$$

Number of iterations: 5

v^0	θ	v^0	θ	v^0	θ	v^0	θ	v^0	θ
0	0.000000	6	0.084306	12	0.169770	18	0.257439	24	0.348161
30	0.442530	36	0.540877	42	0.643290	48	0.749649	54	0.859673
60	0.972968	66	1.089061	72	1.207427	78	1.327516	84	1.448760
90	1.570583	96	1.692407	102	1.813654	108	1.933750	114	2.052126
120	2.168231	126	2.281543	132	2.391589	138	2.497973	144	2.600417
150	2.698801	156	2.793213	162	2.883983	166	2.971706	174	3.057227
180	3.141593	186	3.225958	192	3.311479	198	3.399202	204	3.489972
210	3.584384	216	3.682768	222	3.785212	228	3.891597	234	4.001642
240	4.114954	246	4.231059	252	4.349435	258	4.469531	264	4.590778
270	4.712603	276	4.834426	282	4.955669	288	5.075758	294	5.194125
300	5.310217	306	5.423512	312	5.533537	318	5.639895	324	5.742308
330	5.840656	336	5.935025	342	6.025746	348	6.113415	354	6.198879

Table 6.

Mapping of an ellipse

$$\frac{a}{b} = 1.2$$

Trapezoid rule without correction

Numerical differentiation

$$\theta_i^0 = \frac{2\pi}{60} i$$

Number of iterations: 5

v^0	θ	v^0	θ	v^0	θ	v^0	θ	v^0	θ
0	0.000000	6	0.084302	12	0.169767	18	0.257438	24	0.348160
30	0.442529	36	0.540877	42	0.643290	48	0.749648	54	0.859673
60	0.972968	66	1.089060	72	1.207427	78	1.327516	84	1.448759
90	1.570582	96	1.692407	102	1.813654	108	1.933750	114	2.052126
120	2.168231	126	2.281543	132	2.391589	138	2.497975	144	2.600420
150	2.698806	156	2.793218	162	2.883985	168	2.971704	174	3.057221
180	3.141584	186	3.225951	192	3.311477	198	3.399205	204	3.489978
210	3.584390	216	3.682773	222	3.785215	228	3.891598	234	4.001642
240	4.114953	246	4.231059	252	4.349435	258	4.469531	264	4.590778
270	4.712603	276	4.834426	282	4.955669	288	5.075758	294	5.194124
300	5.310217	306	5.423513	312	5.533539	318	5.639900	324	5.742316
330	5.840667	336	5.935036	342	6.025753	348	6.113414	354	6.198870

It is easily seen that many of the numbers in table 5 and table 6 are very far from the corresponding correct values in table 3.

Nevertheless the differences between the corresponding values in tables 5 and 6 are small. Hence we may conclude:

- I. The correction term in the trapezoid formula must be used.
- II. The results will be nearly the same whether we use numerical differentiation or analytical differentiation to find $\frac{dr(v_i)}{dv}$ and $\frac{d^2r(v_i)}{dv^2}$.

$$\text{ation to find } \frac{dr(v_i)}{dv} \text{ and } \frac{d^2r(v_i)}{dv^2}.$$

The latest results obtained until now are given in table 7. Here the ratio $\frac{a}{b}$ was chosen equal to 2.

Table 7.

Mapping of an ellipse

$$\frac{a}{b} = 2$$

The general method

Analytical differentiation

$$\theta_i^0 = \frac{2\pi}{60} i$$

Number of iterations: 8

v^0	θ	v^0	θ	v^0	θ	v^0	θ	v^0	θ
0	0.000000	6	0.028099	12	0.063557	18	0.111933	24	0.176097
30	0.256444	36	0.351749	42	0.459987	48	0.579089	54	0.706754
60	0.841413	66	0.981431	72	1.125473	78	1.272384	84	1.421138
90	1.570793	96	1.720448	102	1.869202	108	2.016113	114	2.160154
120	2.300171	126	2.434830	132	2.562574	138	2.681594	144	2.789838
150	2.885131	156	2.965473	162	3.029632	168	3.078005	174	3.113471
180	3.141593	186	3.169715	192	3.205180	198	3.253553	204	3.317712
210	3.398054	216	3.493347	222	3.601591	228	3.720611	234	3.848355
240	3.983014	246	4.123031	252	4.267073	258	4.413983	264	4.562737
270	4.712392	276	4.862047	282	5.010801	288	5.157713	294	5.301755
300	5.441772	306	5.576432	312	5.704177	318	5.823198	324	5.931445
330	6.026741	336	6.107088	342	6.171252	348	6.219629	354	6.255086

Some of the values have been checked and the results seem to be accurate.

10. Conclusion.

Provided that the domain which should be mapped onto the unit circle is nearly circular the method seems to be very good. It is fast and accurate. Furthermore the results can be obtained by numerical differentiation if the boundary values of $r = r(v)$ are specified by a table.

In order to find the values of θ in situations when the domain D is not nearly circular, it will often be possible to find a transformation by means of which D can be mapped onto a domain D_1 which is nearly circular, and the method can be used to get the final results ¹⁾.

This problem and other questions in connection with the method will be examined in the nearest future.

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¹⁾ See e.g. the book by Koppenfels und Stallmann [11] or the dictionary [12] by Kober.